Sensitivity via Omitted Variable Bias

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Why sensitivity?

Our workflow so far: pick an ID strategy (e.g. SOO), defend the assumptions, estimate well.

"Defend the assumptions" is the fragile one!

Sensitivity analysis: rather than defending "confounding $= 0$", ask

> "How strong would confounding have to be to overturn the conclusion?"

Old idea — Cornfield et al. (1959) on smoking and lung cancer:

> *... an unobserved confounder ... would need to lead to a nine-fold increase in the odds of smoking to explain away the association ... and it was asserted that such a strong unobserved confounder was very unlikely to exist.*

Warning we'll keep repeating: this is *not* "how much confounding is there."

Effects of violence on attitudes: angry or weary?

Among Darfurian refugees in eastern Chad, did exposure to violence make people on average more willing to make concessions to achieve peace ("weary"), or more set on retribution ("angry")?

Central ID claim: targeting of violence in 2003–2004 was based only on village and gender, otherwise highly indiscriminate. *Being directly harmed is "as-if random within village$\times$gender strata."*

- Aerial bombing was extremely crude and difficult to target
- *Janjaweed* militia attacks targeted on visible characteristics only (notably *female*)
- Purpose was to clear the population, not to pick out rebels

Strategy: SOO conditioning on village and gender.

Darfur: estimate, and the worry

Setup: outcome *PeaceIndex* (0–1); treatment *DirectHarm* (0/1).

$$\mathit{PeaceIndex}_i = \alpha\, \mathit{DirectHarm}_i + \beta^\top \mathit{Village}_i + \phi\, \mathit{female}_i + \epsilon_i.$$

Those directly harmed are on average 9.7 percentage points more pro-peace (SE $=$ 2.3).

But what if the ID strategy is "a little" wrong?

Questions we want sensitivity to help with:

1. If somebody names a confounder I don't have, how strong would it have to be to overturn the result?
2. What about *multiple* confounders, possibly acting non-linearly?
3. How can we report robustness compactly?
4. How can we make informed guesses about plausible confounding?

A landscape, and our choice

Several families of sensitivity analysis exist:

- **Matched-pair Γ bounds** (Rosenbaum 2002): one odds-of-treatment parameter, for matched designs.
- **Parametric** (Imbens 2003; Carnegie–Dorie–Hill 2016): fully specify a model for (D, Y, U), sweep over its parameters.
- **Partial identification** (Manski 1990; monotonicity bounds): assume less, get ranges.
- **OVB-based** (Frank 2000; Hosman–Hansen–Holland 2010; **Cinelli & Hazlett 2020**): use omitted-variable bias as the engine.

We focus on OVB-based: scale-free in R^2, handles multiple/non-linear confounders, and plugs straight into a regression workflow.

A short detour: Frisch–Waugh–Lovell

Suppose we run OLS of Y on D and $\boldsymbol{X}$:

$$Y = \hat{\tau}\, D + \boldsymbol{X}\hat{\beta} + \hat{\epsilon}.$$

The same $\hat{\tau}$ is given by a three-step recipe:

1. Regress Y on $\boldsymbol{X}$, take residuals $Y^{\perp \boldsymbol{X}}$
2. Regress D on $\boldsymbol{X}$, take residuals $D^{\perp \boldsymbol{X}}$
3. Regress $Y^{\perp \boldsymbol{X}}$ on $D^{\perp \boldsymbol{X}}$

In words: *multivariate OLS reduces to univariate OLS, after partialling out* $\boldsymbol{X}$,

$$\hat{\tau} \; = \; \frac{\mathrm{cov}(D^{\perp \boldsymbol{X}},\, Y^{\perp \boldsymbol{X}})}{\mathrm{var}(D^{\perp \boldsymbol{X}})}.$$

This is the central tool for what follows.

Omitted variable bias: setup

Suppose an investigator wishes to run:

$$Y = \hat{\tau}\, D + \boldsymbol{X}\hat{\beta} + \hat{\gamma}\, Z + \hat{\epsilon}_{\text{full}}$$

But Z is unobserved, so they run the *restricted* regression:

$$Y = \hat{\tau}_{\text{res}}\, D + \boldsymbol{X}\hat{\beta}_{\text{res}} + \hat{\epsilon}_{\text{res}}$$

We ask: **how would $\hat{\tau}$ change if you could include Z?**

Whether the “full” regression is wise, or whether its coefficient is the quantity of interest, is a separate question — here we just compute the consequence of omission.

OVB via FWL: $\widehat{\text{bias}} = \hat{\gamma}\hat{\delta}$

Restating FWL,

$$\hat{\tau}_{\text{res}} = \frac{\text{cov}(D^{\perp \boldsymbol{X}},\ Y^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})}.$$

Substitute the (true) full model $Y = \hat{\tau} D + \boldsymbol{X}\hat{\beta} + \hat{\gamma} Z + \hat{\epsilon}_{\text{full}}$:

$$Y^{\perp \boldsymbol{X}} = \hat{\tau}\, D^{\perp \boldsymbol{X}} + \hat{\gamma}\, Z^{\perp \boldsymbol{X}} + \hat{\epsilon}_{\text{full}}^{\perp \boldsymbol{X}}.$$

Then:

$$\hat{\tau}_{\text{res}} = \hat{\tau} + \hat{\gamma} \underbrace{\frac{\text{cov}(D^{\perp \boldsymbol{X}},\ Z^{\perp \boldsymbol{X}})}{\text{var}(D^{\perp \boldsymbol{X}})}}_{=:\ \hat{\delta}} = \hat{\tau} + \hat{\gamma}\hat{\delta}.$$

$\hat{\delta}$ is the coefficient on D in the auxiliary regression

$$Z = \hat{\delta} D + \boldsymbol{X}\hat{\psi} + \hat{\epsilon}_Z.$$

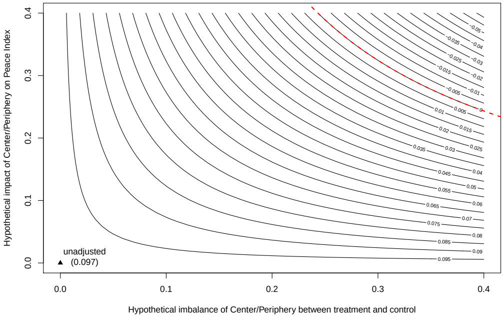
$$\boxed{\widehat{\text{bias}}(\hat{\tau}_{\text{res}}) = \hat{\gamma}\,\hat{\delta} = \text{impact} \times \text{imbalance.}}$$

Consider a confounder

Suppose a colleague says:

> *Those near the center of the village were both more likely to be injured and have different attitudes.*

For each hypothesized pair $(\hat{\gamma}, \hat{\delta})$, the adjusted estimate is $\hat{\tau} = \hat{\tau}_{\text{res}} - \hat{\gamma}\hat{\delta}$.



Why reparameterize

Problems with the "natural" parameterization:

- Units of "impact" and "imbalance" depend on the scale of Z. *Center* happened to be binary, but how would you scale a confounder like "political attitudes"?
- "Name a confounder" is too narrow. We want to handle multiple confounders, possibly acting non-linearly.

Fix: reparameterize using *partial* R^2.

- Scale-free, bounded in $[0, 1]$
- Lets us derive bias *and* variance, do bounds, and handle all confounders jointly.

What is the partial R^2?

The partial correlation is simply correlation *after* partialling out $\boldsymbol{X}$:

$$R_{Y\sim D|\boldsymbol{X}} \;=\; \mathrm{cor}(Y^{\perp \boldsymbol{X}},\, D^{\perp \boldsymbol{X}}).$$

Square it for the partial R^2.

Or, “fraction of variance of D explained by Z beyond what $\boldsymbol{X}$ explains”:

$$R^2_{D\sim Z|\boldsymbol{X}} \;=\; \frac{R^2_{D\sim \boldsymbol{X}+Z} - R^2_{D\sim \boldsymbol{X}}}{1 - R^2_{D\sim \boldsymbol{X}}}.$$

And likewise:

$$R^2_{Y\sim Z|D,\boldsymbol{X}} \;=\; \frac{R^2_{Y\sim D+\boldsymbol{X}+Z} - R^2_{Y\sim D+\boldsymbol{X}}}{1 - R^2_{Y\sim D+\boldsymbol{X}}}.$$

Direction-symmetric: $R^2_{D\sim Z|\boldsymbol{X}} = R^2_{Z\sim D|\boldsymbol{X}}$.

Bias in the R^2 parameterization

Take $\widehat{\text{bias}} = \hat{\gamma}\hat{\delta}$, write each coefficient as cov/var via FWL, replace correlations by partial R^2's, and (after some algebra):

$$\boxed{|\widehat{\text{bias}}| \; = \; \sqrt{\frac{R^2_{Y\sim Z|D,\boldsymbol{X}}\, R^2_{D\sim Z|\boldsymbol{X}}}{1 - R^2_{D\sim Z|\boldsymbol{X}}}} \cdot \frac{\text{sd}(Y^{\perp \boldsymbol{X},D})}{\text{sd}(D^{\perp \boldsymbol{X}})}.}$$

Recall the standard OLS variance fact, $\widehat{\text{se}}(\hat{\tau}_{\text{res}}) = \frac{\text{sd}(Y^{\perp \boldsymbol{X},D})}{\text{sd}(D^{\perp \boldsymbol{X}})\sqrt{\text{df}}}$. So equivalently

$$|\widehat{\text{bias}}| \; = \; \widehat{\text{se}}(\hat{\tau}_{\text{res}}) \sqrt{\frac{R^2_{Y\sim Z|D,\boldsymbol{X}}\, R^2_{D\sim Z|\boldsymbol{X}}}{1 - R^2_{D\sim Z|\boldsymbol{X}}} \cdot \text{df}}\,.$$

Everything data-dependent is now in $\widehat{\text{se}}(\hat{\tau}_{\text{res}})$ and df — both already on your regression printout.

Bias anatomy

Consider the *relative bias* — the fraction of $\hat{\tau}_{\text{res}}$ that is bias. Using $|\hat{\tau}_{\text{res}}| = |t_D| \,\widehat{\text{se}}(\hat{\tau}_{\text{res}})$,

$$\frac{|\widehat{\text{bias}}|}{|\hat{\tau}_{\text{res}}|} = \frac{\sqrt{\text{df}} \cdot \text{BF}}{|t_D|} = \frac{\text{BF}}{|f_{Y\sim D|\boldsymbol{X}}|},$$

where $\text{BF} := \sqrt{\frac{R^2_{Y\sim Z|D,\boldsymbol{X}} R^2_{D\sim Z|\boldsymbol{X}}}{1-R^2_{D\sim Z|\boldsymbol{X}}}}$ is the "bias factor" and $f_{Y\sim D|\boldsymbol{X}} := R_{Y\sim D|\boldsymbol{X}} / \sqrt{1 - R^2_{Y\sim D|\boldsymbol{X}}}$ is the partial Cohen's f. Three things to take away:

- BF is the expression of confounder strength.
- $f_{Y\sim D|\boldsymbol{X}}$ — essentially how much of Y's variance the treatment uniquely explains — is *all* that enters from the data. It is what confounding is competing to explain.
- $f_{Y\sim D|\boldsymbol{X}}$ is *not* a function of sample size: more data does not help you fight confounding.

Variance anatomy

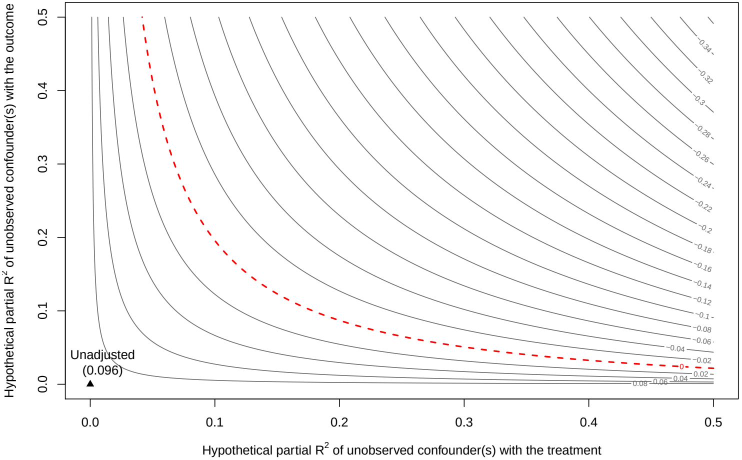
Applying the same OLS variance fact to the *full* regression and combining:

$$\widehat{\text{se}}(\hat{\tau}) \;=\; \widehat{\text{se}}(\hat{\tau}_{\text{res}}) \cdot \sqrt{\underbrace{\left(1 - R^2_{Y\sim Z|\boldsymbol{X},D}\right)}_{\text{VRF}} \underbrace{\frac{1}{1 - R^2_{D\sim Z|\boldsymbol{X}}}}_{\text{VIF}} \underbrace{\frac{\text{df}}{\text{df}-1}}_{\text{df adj.}}}.$$

- **VRF**: had we included Z, residual Y-variance would shrink.
- **VIF**: had we included Z, its correlation with D would inflate the SE.

First, good to have this in your head

Figure: Contour plot in the partial R^2 parameterization



Two simple statistics for routine reporting

Robustness Value (RV): the level of (equal) partial-R^2 association with D and Y that would eliminate the effect.

- RV_q: the equal-association level needed to bring the estimate down by proportion q
- RV_α: the equal-association level needed to bring the t-stat to the threshold for level α

Partial R^2 of D with Y: $R^2_{Y \sim D|\mathbf{X}}$

- If confounding explained *all* residual Y-variance, this is how much of D (residual variance) it would have to explain to bring the estimate to zero.

Both come from t_D and df alone — you can compute them from a published regression table.

Augmented regression table: Darfur

Outcome: *Peace Index*

Treatment:	Est.	SE	t	$R^2_{Y\sim D\|\boldsymbol{X}}$	RV	$\text{RV}_{\alpha=0.05}$
Directly Harmed	0.097	0.023	4.18	2.2%	13.9%	7.6%
df $=$ 783		*Bound (1$\times$ female): $R^2_y = 12\%$, $R^2_d = 1\%$*				

Somebody tell us:

- what does this RV mean?
- what does this $\text{RV}_{\alpha=0.05}$ mean?
- what does this $R^2_{Y\sim D|\boldsymbol{X}}$ mean?

Bounding by comparison to observed covariates

One good interpretational wedge: comparing to an important observed covariate.

For an observed covariate X_j, define:

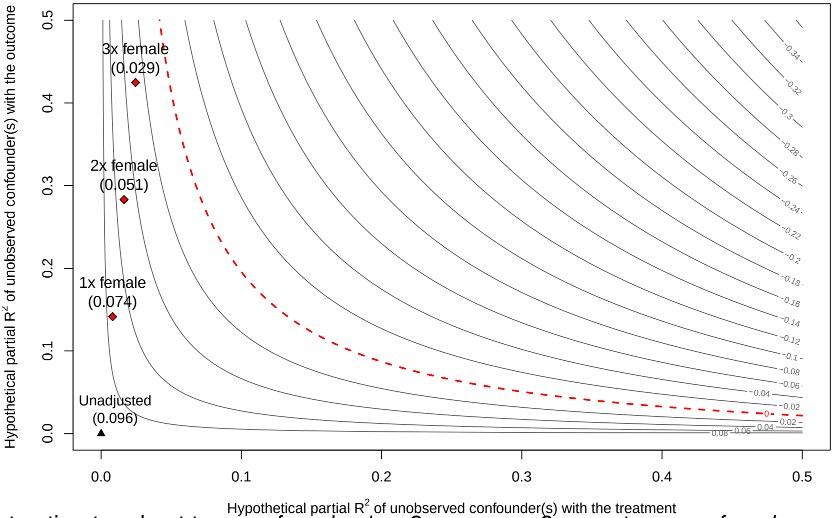
$$k_D := \frac{R^2_{D\sim Z|\boldsymbol{X}_{-j}}}{R^2_{D\sim X_j|\boldsymbol{X}_{-j}}}, \qquad k_Y := \frac{R^2_{Y\sim Z|\boldsymbol{X}_{-j},D}}{R^2_{Y\sim X_j|\boldsymbol{X}_{-j},D}}.$$

"Z is at most k_D times as predictive of D, and k_Y times as predictive of Y, as X_j is." Given $\{X_j, k_D, k_Y\}$, this *bounds* both $R^2_{Y\sim Z|D,\boldsymbol{X}}$ and $R^2_{D\sim Z|\boldsymbol{X}}$, and hence the worst-case bias. Conservative for multiple, non-linear confounders.

This shows the implications of an assumption — you still have to argue the assumption itself is believable.

Example: within Darfur villages, hard to imagine targeting on any basis stronger than *female*. So use $X_j = \textit{female}$, and vary k.

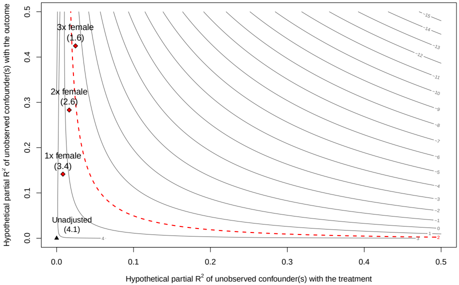
Contour plot with bounds: Darfur



Point estimate robust to a confounder 1×, 2×, or even 3× as strong as *female*.

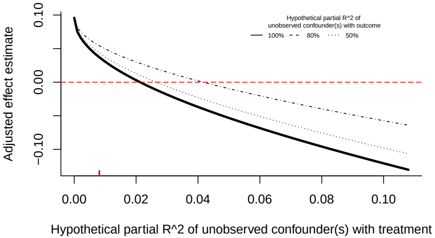
Inference and extreme scenarios

t-statistic contour



t stays significant against a confounder up to $\sim 2\times$ *female*; loses significance by $3\times$.

Extreme scenario



Assume confounding explains *all* residual outcome variance ($R^2_{Y\sim Z|\boldsymbol{X},D} = 1$). Z still needs to explain $\geq 2.2\%$ of D to zero the effect. You pay a real price when you can't bound.

How to argue it (and what readers should demand)

Defending a result. Sometimes you can argue from design and context, e.g. for Darfur:

> *"...aircraft dropped makeshift and unguided bombs over villages, and militia raided without concern for who they would attack — the only known major exception, due to sexual assaults, was targeting gender, which is also one of the most visually apparent characteristics.* ***A confounder twice as strong as female would be surprising.****"*

Other times you can't, and the analysis simply reveals that, which is useful.

Critiquing a result. Dismissing a finding requires arguing for confounding of a specific strength, not just any confounder. (t and df alone get you started; as a reader/reviewer you have the info.)

The reframe. This is never "how much confounding is there" — it's "how much would it take to overturn the conclusion." No magic RV threshold; the point is to refine the debate, not hide behind a number. Whether the answer is reassuring or unsettling, it's the honest conversation.

Resources

Papers (all linked from chadhazlett.com/publications):

- Cinelli & Hazlett (2020). *Making sense of sensitivity: Extending omitted variable bias.* JRSS B. — the framework
- Hazlett & Parente (2023). *From "Is it unconfounded?" to "How much confounding would it take?"* JoP. — applied / how-to-argue companion
- Cinelli, Ferwerda & Hazlett (2024). *sensemakr: Sensitivity Analysis Tools for OLS in R and Stata.* Observational Studies. — software paper
- *Wildfire Exposure Increases Pro-Climate Political Behaviors* — worked applied example

Software & tutorials:

- `sensemakr` R/Stata + worked vignettes: carloscinelli.com/sensemakr
- Carlos Cinelli's site (talks, tutorials, videos): carloscinelli.com